Lecture 8: Likelihood Ratio Tests

IEOR E4709: Statistic Analysis and Time Series
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1 Hypothesis Testing

Consider independent and identically distributed samples $X_1, X_2, \ldots, X_n$ coming from a Gaussian distribution $N(\mu, \sigma^2)$ with unknown mean μ and unknown standard deviation σ. We want to know whether $\mu = \mu_0$ for some pre-specified mean μ_0. For instance, if X_i's are returns of a stock observed over a period of time, we may be interested in testing whether the returns of this stock have zero mean. In this case, we would take $\mu_0 = 0$. Let us consider the sample mean estimator $\bar{X} = \frac{1}{n}\sum_{i=1}^n X_i$. This is an unbiased estimator of μ. Would you reject the hypothesis when $\bar{X}$ is close to zero? Intuitively, we tend not to reject in this case. For instance, suppose that $|\bar{X} - \mu_0| \leq c\frac{\hat{\sigma}_s}{\sqrt{n}}$, where $\hat{\sigma}_s = \sqrt{\frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X})^2}$ is the sample standard deviation and c is a chosen constant. Then we would not reject the hypothesis, because $\bar{X}$ is sufficiently close to μ_0. We would reject, instead, if this inequality is not satisfied.

Question: What is the criterion used? What is the probability of *false negatives*, that is of wrongly not rejecting the hypothesis? What is the probability of *false positives*, that is of wrongly rejecting the hypothesis? The probability of wrongly rejecting the hypothesis is

$$\mathbb{P}_{\mu_0}\left(|\bar{X} - \mu_0| > c\frac{\hat{\sigma}_s}{\sqrt{n}}\right) = \mathbb{P}_{\mu_0}\left(\frac{|\bar{X} - \mu_0|}{\frac{\hat{\sigma}_s}{\sqrt{n}}} > c\right) = 2T_{n-1}(-c),$$

The ratio term inside $\mathbb{P}_{\mu_0}$ is a random variable, whose distribution is the student's t distribution with $n-1$ degrees of freedom (we lose one degree of freedom because we have replaced the standard deviation of the Gaussian with its sample estimator). We denote the CDF of a Student's t distribution with n degree of freedom by T_n.

The probability of wrongly not rejecting the hypothesis is the maximum, over all values $\mu \neq \mu_0$,

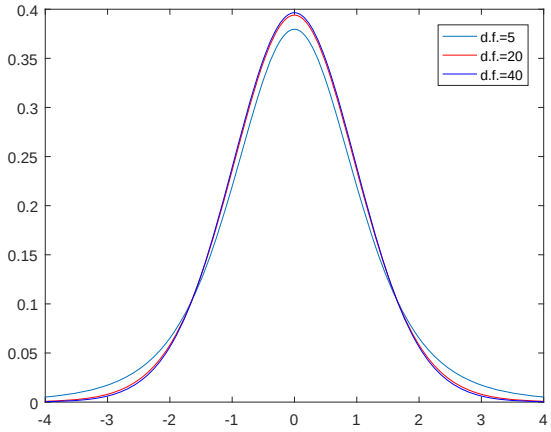


Figure 1: The probability density function of the t-distribution with 5, 20, and 40 degrees of freedom, respectively.

of the following probabilities:

$$\begin{aligned}
\mathbb{P}_\mu\left(|\bar{X}-\mu_0| < c\frac{\hat{\sigma}_s}{\sqrt{n}}\right) &= \mathbb{P}_\mu\left(-c\frac{\hat{\sigma}_s}{\sqrt{n}} \le \bar{X}-\mu_0 \le c\frac{\hat{\sigma}_s}{\sqrt{n}}\right) \\
&= \mathbb{P}_\mu\left(-c \le \frac{\bar{X}-\mu_0}{\frac{\hat{\sigma}_s}{\sqrt{n}}} \le c\right) \\
&= \mathbb{P}_\mu\left(-c-\frac{\mu-\mu_0}{\hat{\sigma}_s/\sqrt{n}} \le \frac{\bar{X}-\mu}{\frac{\hat{\sigma}_s}{\sqrt{n}}} \le c-\frac{\mu-\mu_0}{\hat{\sigma}_s/\sqrt{n}}\right) \\
&= T_{n-1}\left(c-\frac{\mu-\mu_0}{\hat{\sigma}_s/\sqrt{n}}\right) - T_{n-1}\left(-c-\frac{\mu-\mu_0}{\hat{\sigma}_s/\sqrt{n}}\right)
\end{aligned}$$

We would like both the probability of wrongly rejecting the hypothesis, and the probability of wrongly not rejecting the hypothesis, to be small.

Let us now try to formalize all this at a more general level. Assume that the parameter θ belongs to the space Θ. Consider the *null hypothesis* $H_0 : \theta \in \Theta_0$, versus the *alternative hypothesis* $H_a : \theta \notin \Theta_0$, that is $H_a : \theta \in \Theta \setminus \Theta_0$. Suppose we obtain samples $X_1, X_2, \ldots, X_n$. The rejection region will be a subset R of $\mathbb{R}^n$: we reject H_0 if $X = X_1, \ldots, X_n \in R$, and do not reject otherwise. We have two type of errors which can occur:

- **Type I error:** This is also known as a *false positive*: the error of rejecting a null hypothesis when it is actually true. In plain words, this occurs when we are observing a difference even if in truth there is no difference. Formally, this is $\sup_{\theta\in\Theta_0} \mathbb{P}_\theta(X \in R)$
- **Type II error:** This is also known as a *false negative*: the error of not rejecting a null hypothesis when the alternative hypothesis is the true state of nature. In plain words, this

occurs when we are failing to observe a difference even if in reality there is one. Formally, this is $\sup_{\theta\in\Theta\setminus\Theta_0}\mathbb{P}_\theta(X\notin R) = 1-\inf_{\theta\in\Theta\setminus\Theta_0}\mathbb{P}_\theta(X\in R)$.

We would like to minimize both Type I and Type II errors simultaneously. Usually, we control for the Type I error first. Notice that null and alternative hypothesis are not symmetric: not rejecting H_0 is different from accepting H_0.

Let α be the probability of committing the error to reject the null hypothesis H_0. α is called the *significance level* of the test. Mathematically, this is equal to $\sup_{\theta\in\Theta_0}\mathbb{P}_\theta(X\in R)$. Usually, when we perform a test, we specify a significance level. The rejection region is usually specified via a **test statistic**, which is a function of the sample X. Given a test statistic $W(X)$, we define the rejection region

$$R := \{x : W(x) > c\},$$

where c is a given threshold. Such threshold c will be chosen based on the significance level of the test. For this reason, it will be important to know the distribution $W(X)$ under the null hypothesis H_0.

2 Statistical Tests

There are two types of statistical tests: One-tailed tests and two-tailed tests. Both are ways of computing the statistical significance of a parameter inferred from a data set, in terms of a test statistics. A two-tailed test is used if deviations of the estimated parameter in either direction from some benchmark value are considered theoretically possible; in contrast, a one-tailed test is used if only deviations in one direction are considered possible.

2.1 One-Side Hypotheses

Let us first focus on one-side hypotheses. Consider i.i.d. samples $X_1, X_2, \ldots, X_n$, coming from a Gaussian distribution with unknown mean μ and variance σ^2.

Objective: Test the hypothesis $H_0 : \mu \leq \mu_0$ versus the alternative hypothesis $H_a : \mu > \mu_0$. Consider the test statistic

$$W(X) = \frac{\bar{X}-\mu_0}{\hat{\sigma}_s/\sqrt{n}},$$

where $\bar{X}$ is the sample mean and $\hat{\sigma}_s$ is the sample (unbiased) standard deviation estimator. We reject the null hypothesis if and only $W(X) > c$ for some constant c. Under the Gaussian probability distribution with mean μ and variance σ^2, the random variable $(\bar{X}-\mu)/(\hat{\sigma}_s/\sqrt{n})$ follows a Student's t distribution with $n-1$ degrees of freedom. Therefore,

$$\mathbb{P}_\mu\left(W(X) > c\right) = \mathbb{P}_\mu\left(\frac{\bar{X}-\mu}{\hat{\sigma}_s/\sqrt{n}} > c - \frac{\mu-\mu_0}{\hat{\sigma}_s/\sqrt{n}}\right) = 1 - T_{n-1}\left(c - \frac{\mu-\mu_0}{\hat{\sigma}_s/\sqrt{n}}\right).$$

We can then calculate the **type I error**. This is given by

$$\sup_{\mu\leq\mu_0}\mathbb{P}_\mu\left(W(X) > c\right) = 1 - T_{n-1}(c).$$

Suppose we want significance level α, e.g. 5%. Then we can set $c = T_{n-1,1-\alpha}$, where c is the $1-\alpha$-quantile of the distribution, i.e., $\mathbb{P}(V > c) = \alpha$ if the random variable V follows a Student's t distribution with $n-1$ degrees of freedom. Hence, we would reject the hypothesis H_0 if and only if

$$\bar{X} > \mu_0 + T_{n-1,\alpha}\hat{\sigma}_s/\sqrt{n}.$$

3 Two Simple Hypotheses

Consider two *simple* hypotheses, that is $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$. Define $\beta = \mathbb{P}(X \notin R \,|H_1)$, i.e., the probability of failing to reject the null hypothesis, given that the alternative hypothesis holds (here, R denotes the rejection region). The *power* of the test is $1-\beta$. Usually we fix the type I error probability $\alpha = \mathbb{P}(reject\ H_0|H_0)$ also known as the size of the test (typical values of α are $\alpha = 0.05, 0.01$, etc...) and we look for a test which minimizes the probability β of a type II error: this is called a *most powerful* test of *size* α. Intuitively, we reject H_0 in favor of H_1 if the likelihood of θ_1 is much larger than the likelihood of θ_0 given the data.

4 Neyman-Pearson Lemma

To test two simple hypotheses, we use likelihood ratio tests inspired by the following lemma.

Lemma 4.1 (**Neyman-Pearson Lemma**). *Let $X = (X_1, X_2, \ldots, X_n)$ be a random sample from a distribution with parameter $\theta \in \{\theta_0, \theta_1\}$, and let $L(\theta; x)$ be the likelihood function. Suppose there exists a test at significance level α such that the test rejects the null hypothesis if*

$$\frac{L(\theta_1; x)}{L(\theta_0; x)} > k,$$

where the constant k is defined so that $\mathbb{P}_{\theta_0}\left(\frac{L(\theta_1;X)}{L(\theta_0;X)} > k\right) = \alpha$. Then, this test is most powerful at significance level α for testing the null hypothesis $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$. This test is called the likelihood ratio test.

Example: Suppose that we have a random sample $X_1, X_2, \ldots, X_n$ following a Gaussian distribution with unknown mean μ and known variance σ^2. Let μ_0, μ_1 be given, and assume that $\mu_1 > \mu_0$. We would like to test $H_0 : \mu = \mu_0$ against $H_1 : \mu = \mu_1$. Using the Neyman-Pearson Lemma, the best test is a likelihood ratio test. To calculate the test statistics, let us use $L(\mu_1|x)$ to denote the Gaussian likelihood with mean μ_1 and variance σ^2. Let us also use $L(\mu_0|x)$ to denote the Gaussian likelihood with mean μ_0 and variance σ^2. Notice that

$$\frac{L(\mu_1|x)}{L(\mu_0|x)} \geq k$$

if and only if

$$LL(\mu_1|x) - LL(\mu_0|x) \geq \log(k),$$

whic holds if and only if

$$-\sum_i \left[(x_i - \mu_1)^2 - (x_i - \mu_0)^2\right] \geq 2\sigma^2 \log(k),$$

which in turn holds if and only if

$$(\mu_1 - \mu_0)\bar{x} \geq k',$$

which is true if and only if $\bar{x} \geq c$, where $c = \frac{k'}{\mu_1 - \mu_0}$ is a constant (i.e., independent of $x = (x_1, x_2, \ldots, x_n)$), and $\bar{x} = \frac{1}{n}\sum_i x_i$. Hence, for a size α test, we choose c so that $\mathbb{P}_{\mu_0}(\bar{X} \geq c) = \alpha$, or equivalently such that

$$\mathbb{P}_{\mu_0}\left(\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \geq \frac{c - \mu_0}{\sigma/\sqrt{n}}\right) = \alpha.$$

Hence, we want

$$\frac{c - \mu_0}{\sigma/\sqrt{n}} = z_{1-\alpha},$$

where $P(Z \geq z_{1-\alpha}) = P(Z \leq z_\alpha) = \Phi(z_\alpha) = \alpha$, and Φ is the standard Gaussian cumulative distribution function, i.e. $c = \mu_0 + \sigma z_{1-\alpha}/\sqrt{n}$. So, the most powerful test of H_0 versus H_1 at level α becomes "reject H_0 if and only if $\bar{X} \geq \mu_0 + \sigma\frac{z_{1-\alpha}}{\sqrt{n}}$".